Name		Team	

The visit times

Comet: 33, 34, 35, 36, 37, 38, 39, 40, 40, 41, 42, 43, 44.

Nebula: 26, 27, 28, 29, 30, 31, 32, 33, 35, 38.

Comet has thirteen times and twelve of them are different from each other. A column graph of those raw numbers would have twelve bars, nearly all of them one high.

Part A · Counting into groups

SF 1

Count Comet’s thirteen visits into these groups. Then do Nebula’s ten. Check each column adds to the right total.

Group (seconds)	Comet	Nebula
26 to 29		
30 to 33		
34 to 37		
38 to 41		
42 to 45		
Total		

Part B · Making the groups work

CF 3

Every group above is four seconds wide. Why do all the groups have to be the same width? What would go wrong if one were twice as wide?

CF 4

A team uses the groups 30–35 and 35–40. What is wrong with those two? Write a pair that fixes it.

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CF 5 Try groups of 2 seconds, then groups of 10. What goes wrong with each? About how many groups seems to work best?

CF 6 Describe the shape of the two grouped distributions in one sentence that compares them.

Part C · What grouping costs

CU 7 Once the times are grouped, 38 seconds and 41 seconds look exactly the same. What have you lost by grouping, and why is it still worth doing?

CU 8 Comet had 13 times and Nebula had 10. Does that change how you should compare the two grouped displays? Say what you would do about it.

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Our own times

How many usable times		Shortest / longest	

Fewer than about eight? Use the ruler drop data on the spreadsheet instead, and write that in your decision log.

Our groups

Choose your own width. Write the From and To for each group, then count. The total must match the number of times you actually have.

From	To	How many	Width check

Our total	

A sketch of our display

Bars touching, because the groups run into each other. Title it and label both axes.

Describing it

THE SHAPE OF OUR GROUPED TIMES

WHAT THE GROUPING HAS HIDDEN

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Exit ticket

On your own. Five minutes.

SF 1

Twelve measurements: 21, 24, 22, 29, 31, 26, 33, 25, 28, 30, 23, 27. Put them into the groups below. Check your counts add to 12.

Group	21 to 25	26 to 30	31 to 35	Total
How many				

CF 2

A team uses the groups 20–25, 25–30 and 30–35. Name the problem and write a set of groups that fixes it.

CU 3

Someone says "grouping the data is cheating, because you are hiding the real numbers". Write a reply.

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